

objective = loss + $\Omega(\text{new tree})$



Taylor: second-order approximation
with gradients g_i and Hessians h_i



regroup the rows by leaf: G_j, H_j



set the derivative to 0



leaf output

$$w_j = -\frac{G_j}{H_j + \lambda}$$



put w_j back in

similarity score

$$\frac{G_j^2}{H_j + \lambda}, \text{ and the gain}$$